Rayapati Naga Vamsi Krishna

me23btech11052@iith.ac.in | vamsikrishna020705@gmail.com | +91 8121147348 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.83
Sasi Junior College	Class XII (BIEAP)	2023	98.40%
Sasi English Medium High School	Class X (BSEAP)	2021	97.17%

WORK EXPERIENCE

AI Developer Intern | Diagonal Matrix

- Built and tested APIs using AI-assisted coding agents (**Codex**) and integrated Microsoft services--**Outlook, Excel, and Dynamics 365**--into **JAI, an agentic AI platform** that retrieves data from existing files and databases based on natural-language user prompts, **eliminating manual searching across spreadsheets and databases for retail teams**.
- Designed and implemented **20+ automated workflows** across sales, marketing, customer service, and inventory domains for **retail industry clients** using **Langflow**, with **Dynamics 365** as the underlying data platform -- **reducing manual data-retrieval time across manager, executive, and operational staff levels**.
- Integrated the **Dynamics 365 MCP server** to execute and validate CRUD operations, verifying data consistency through the Dynamics UI and successfully demonstrating workflow reliability.

PROJECTS

OrderFlow | Event-Driven Microservices E-Commerce Backend

- Architected **5 microservices** (Auth, Product, Order, Inventory, Payment) using **Java 21** and **Spring Boot**, with **API Gateway**, **Eureka** service discovery, and **database-per-service PostgreSQL**.
- Built an **asynchronous, event-driven order pipeline** using **Kafka** spanning order creation, inventory reservation, and payment, with **idempotent event handling** ensuring reliable, duplicate-safe processing.
- Implemented **JWT authentication** and **Redis caching**, **reducing average response latency by 33% (7.0ms 4.7ms on cache hits, benchmarked over 100 requests)**; integrated **Stripe** test-mode payments and documented APIs with **Swagger/OpenAPI**.
- Containerized all 5 microservices and supporting infrastructure (**PostgreSQL, Redis, Redpanda, Eureka, API Gateway**) using **Docker** and **Docker Compose**, enabling **one-command deployment** and consistent local environments.

Distributed Rate Limiter

- Built a **distributed rate limiter** using **Spring Boot** and **Redis**, supporting **Fixed Window** and **Token Bucket** algorithms for controlled, reliable, and scalable API traffic management.
- Implemented atomic **Redis Lua scripts** for request validation, counter updates, and token refilling, ensuring consistent, race-condition-free rate limiting under high concurrent request loads.
- **Benchmarked the fixed-window limiter under 1,000 concurrent requests, sustaining 718 req/sec while correctly enforcing a strict 100-request limit and rejecting the remaining 900** -- validating race-condition-free behavior under load.
- Integrated **Micrometer** and **Spring Boot Actuator** to monitor request throughput, rejection rates, and Redis operation latency.

PolyCode | AI-Powered Coding Platform

- Built an **AI-driven problem generation and validation pipeline** using Groq LLM APIs, generating **100+ validated problems** with duplicate detection, defensive parsing, and automatic retry mechanisms for malformed output.
- Developed an **automated validation system** that compiles and executes AI-generated reference code, generates diverse edge-case test cases, and derives accurate expected outputs automatically.
- Implemented and **deployed a live online judge** supporting **Java and C++** with verdict evaluation, hidden test cases, AI-powered submission review, and detailed complexity analysis feedback.
- Developed a relational data model using **Spring Data JPA** and **Hibernate** to manage problems, examples, hidden test cases, and submissions with structured entity relationships and persistent evaluation results.

SKILLS

Languages: Java, Python, JavaScript, SQL

Backend: Spring Boot, Spring Security, REST APIs, Microservices

Databases: MySQL, PostgreSQL, Redis

Frontend: React

System Design: API Gateway, Service Discovery, Kafka, Event-Driven Architecture, Caching, Rate Limiting

Tools & Testing: Git, Docker, JUnit, Mockito, Swagger

Core: DSA, OOPs, DBMS, OS, Computer Networks

ACHIEVEMENTS

- Solved **800+ problems** on LeetCode, achieving a maximum contest rating of **1777** (top **9.34%** globally).
- Achieved **rank 544** in LeetCode Weekly Contest 510 and **rank 651** in Weekly Contest 505.

POSITION OF RESPONSIBILITY

Hospitality Head, Milan

- Led a 15-member hospitality team for the Inter-Hostel Competition, coordinating logistics and event execution.

Student Mentor, Sunshine

- Guided 20 first-year students through their academic and social transition into college life.

Hospitality Coordinator, Inter IIT 2025

- Managed logistics and coordination for a multi-institute national-level event.

Core Member, Aero Club

- Built and worked on RC planes, contributing to the club's design and flight-testing activities.